

Chapter 5A: Algebra

8 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-ALG-C-01 generalises number properties to operate with algebraic expressions.



5.01 The commutative and associative laws

5.02 The distributive law

5.03 Variables and algebraic notation

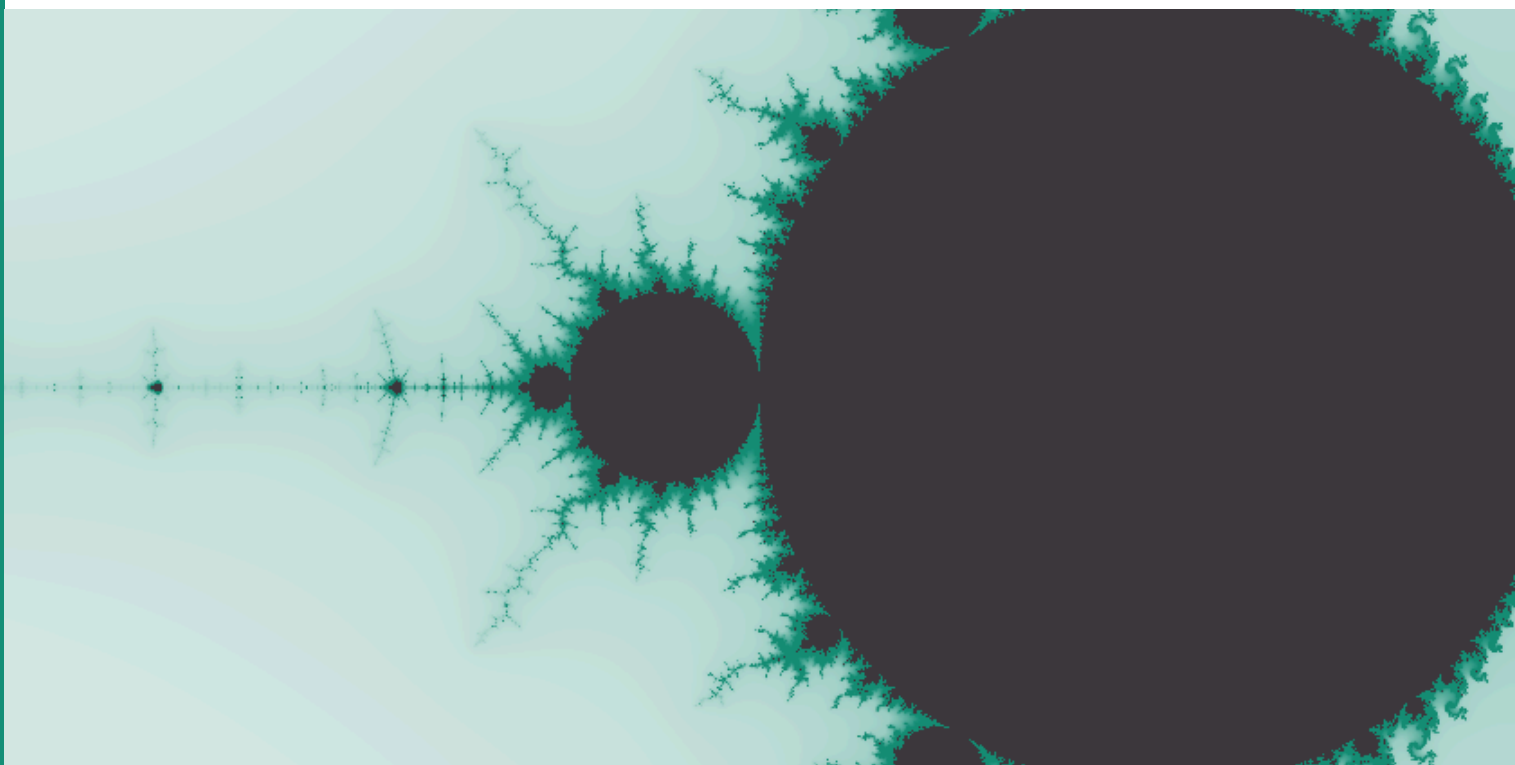
5.04 Expressions from words

5.05 Substitution

5.07 Adding and subtracting terms

5.08 Multiplying terms

5.09 Dividing terms



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

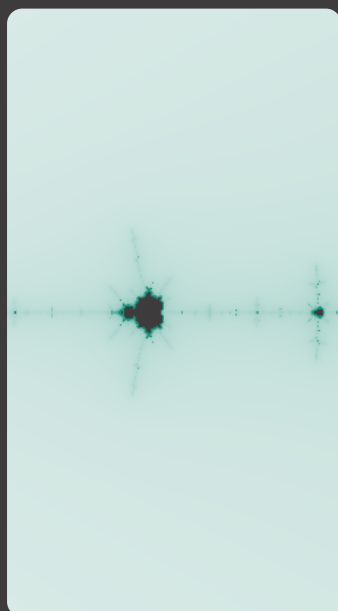
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
5.01	The commutative and associative laws	3		☐
5.02	The distributive law	7		☐
5.03	Variables and algebraic notation	11		☐
5.04	Expressions from words	15		☐
5.05	Substitution	19		☐
5.07	Adding and subtracting terms	23		☐
5.08	Multiplying terms	27		☐
5.09	Dividing terms	31		☐

In this chapter you will

- use the commutative, associative and distributive laws
- use variables and algebraic notation
- write expressions from words
- substitute into expressions and formulas
- add, subtract, multiply and divide algebraic terms



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
5.01

The commutative and associative laws

Date _____

Learning intentions

We are learning to:

- use the commutative and associative laws to make calculations easier

Success criteria

I can:

- ☐ know that order does not matter for + and $\times$ (commutative law)
- ☐ know that grouping does not matter for + and $\times$ (associative law)
- ☐ regroup numbers to make friendly pairs such as 10s and 100s

Key terms Write the definition as your teacher explains it.

Commutative

Associative

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Evaluate by pairing friendly numbers.

a $7 + 18 + 3$

b $91 + 18 + 9$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Evaluate by pairing friendly numbers.

14 QUESTIONS

REMEMBER Commutative: $a + b = b + a$ and $a \times b = b \times a$.

1 $7 + 18 + 3$

2 $91 + 18 + 9$

3 $46 + 25 + 4$

4 $15 + 38 + 5$

5 $64 + 17 + 36$

6 $8 + 29 + 12 + 1$

Questions 7–14: pair friendly numbers.

7 $6 + 27 + 4$

8 $55 + 19 + 45$

9 $2 \times 43 \times 5$

10 $4 \times 9 \times 25$

11 $8 + 71 + 2$

12 $20 \times 7 \times 5$

13 $33 + 48 + 7$

14 $5 \times 13 \times 2$

YOU DO

On your own: **Easy** Set B. Evaluate by pairing friendly numbers.

6 QUESTIONS

1 $2 \times 17 \times 5$

4 $4 \times 13 \times 25$

2 $25 \times 31 \times 4$

5 $50 \times 7 \times 2$

3 $5 \times 9 \times 20$

6 $2 \times 8 \times 5 \times 5$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Evaluate by pairing friendly numbers.

a $2 \times 17 \times 5$

Your teacher works this out. Copy every step.

b $25 \times 31 \times 4$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Evaluate mentally.

6 QUESTIONS

REMEMBER Associative: $(a + b) + c = a + (b + c)$.

1 13×30

3 12×50

5 600×4

2 7×400

4 30×80

6 25×12

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Evaluate mentally.

a 13×30

Your teacher works this out. Copy every step.

b 7×400

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER Look for pairs making 10, 100 or 1000.

- 1** Does the commutative law work for subtraction? Test it with $10 - 4$.

- 3** Evaluate $125 \times 17 \times 8$ mentally.

- 2** Does the associative law work for division? Test it with $(24 \div 6) \div 2$.

- 4** Evaluate $1 + 2 + 3 + \dots + 20$ by pairing numbers.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Evaluate $16 + 27 + 4$.

Level 2

Evaluate $5 \times 23 \times 2$.

Level 3

Evaluate $20 \times 13 \times 5$.

Extension · Stretch yourself**Worked example**Evaluate $4 \times 7 \times 25 \times 3 \times 5 \times 2$ mentally.

1. Regroup: $(4 \times 25) \times (5 \times 2) \times 7 \times 3$
2. $100 \times 10 \times 21$

Answer: 21 000

YOU DO

On your own: read the worked example, then try these.

- E1**
- Evaluate
- $8 \times 125 \times 9 \times 5 \times 2$
- mentally.

- E2**
- Explain why
- $38 + 47 + 62 + 53$
- is easy to do mentally.

Lesson summary

Missed the lesson? Start here.

Remember

1. Commutative: $a + b = b + a$ and $a \times b = b \times a$.
2. Associative: $(a + b) + c = a + (b + c)$.
3. Look for pairs making 10, 100 or 1000.
4. Not true for $-$ and $\div$.

Level 1

$7 + 18 + 3$

1. $(7 + 3) + 18$

Answer: 28

Level 2

$25 \times 31 \times 4$

1. $(25 \times 4) \times 31$

Answer: 3100

Level 3

13×30

1. $13 \times 3 \times 10$

Answer: 390

Extension

$4 \times 7 \times 25 \times 3 \times 5 \times 2$

1. $100 \times 10 \times 21$

Answer: 21 000

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.02

The distributive law

Date _____

Learning intentions

We are learning to:

- use the distributive law for mental arithmetic

Success criteria

I can:

- ☐ split a number into tens and units, or a friendly number $\pm$ a bit
- ☐ multiply each part, then add or subtract
- ☐ choose the easiest way to split

Key terms Write the definition as your teacher explains it.

Distributive law

Expand

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the distributive law.

a 22×9 as $22 \times (10 - 1)$

b 17×8 as $(10 + 7) \times 8$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Complete the working to evaluate.

14 QUESTIONS

REMEMBER $a \times (b + c) = a \times b + a \times c$.

1 $14 \times 6 = 10 \times 6 + 4 \times 6$

2 $23 \times 4 = 20 \times 4 + 3 \times 4$

3 $31 \times 7 = 30 \times 7 + 1 \times 7$

4 $19 \times 5 = 20 \times 5 - 1 \times 5$

5 $42 \times 3 = 40 \times 3 + 2 \times 3$

6 $29 \times 4 = 30 \times 4 - 1 \times 4$

Questions 7–14: use the distributive law.

7 16×5

8 21×6

9 19×4

10 32×3

11 99×7

12 41×5

13 18×9

14 102×4

YOU DO

On your own: **Easy** Set B. Use the distributive law to evaluate.

6 QUESTIONS

1 22×9

4 13×12

2 17×8

5 15×7

3 27×11

6 39×3



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Use the distributive law.

a 27×11

Your teacher works this out. Copy every step.

b 13×12

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Use the distributive law. Show your working.

6 QUESTIONS

REMEMBER Split into tens and units: $17 = 10 + 7$.

1 48×6

3 101×23

5 8×199

2 99×35

4 53×12

6 65×21

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.**Example 3** Use the distributive law.

a 48×6

Your teacher works this out. Copy every step.

b 99×35

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER Or near a friendly number: $9 = 10 - 1$.**1** Evaluate $7 \times 68 + 7 \times 32$ mentally.**3** Show two different ways to evaluate 18×25 .**2** Evaluate $15 \times 43 - 15 \times 33$ mentally.**4** Explain why $6 \times 97 = 6 \times 100 - 6 \times 3$.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Evaluate 12×7 using the distributive law.**Level 2**Evaluate 19×6 .**Level 3**Evaluate 102×45 .

 Extension · Stretch yourself Worked example

Evaluate 999×47 mentally.

1. $999 = 1000 - 1$
2. $47\ 000 - 47$

Answer: 46 953

YOU DO

On your own: read the worked example, then try these.

- E1** Evaluate 1001×64 mentally.

- E2** Evaluate $37 \times 99 + 37$ mentally.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. $a \times (b + c) = a \times b + a \times c$.
 3. Or near a friendly number: $9 = 10 - 1$.

2. Split into tens and units: $17 = 10 + 7$.
4. Multiply each part, then add or subtract.

 Level 1

$$22 \times 9$$

1. $22 \times 10 - 22$

Answer: 198

 Level 2

$$17 \times 8$$

1. $80 + 56$

Answer: 136

 Level 3

$$99 \times 35$$

1. $3500 - 35$

Answer: 3465

 Extension

$$999 \times 47$$

1. $47\ 000 - 47$

Answer: 46 953

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LESSON
5.03

Variables and algebraic notation

Date _____

Learning intentions

We are learning to:

- use a variable to describe a pattern
- write and read algebraic notation

Success criteria

I can:

- ☐ write a general rule using a variable
- ☐ simplify notation, e.g. $5 \times a \times 2 = 10a$
- ☐ write a term in expanded form

Key terms Write the definition as your teacher explains it.

Variable

Pronumeral

Term

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write the rule, then simplify.

a Write the rule for: $3 - (-3) = 2 \times 3$, $5 - (-5) = 2 \times 5$,
 $11 - (-11) = 2 \times 11$

Your teacher works this out. Copy every step.

b Simplify $x + x + x + x$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Simplify.

14 QUESTIONS

REMEMBER A **VARIABLE** is a letter that stands for a number.

1 $x + x + x + x$

2 $u \div 5$

3 $2y - y$

4 $5 \times a \times 2$

5 $m \times 7$

6 $p \times q$

Questions 7–14: simplify.

7 $a + a + a$

8 $4 \times b$

9 $y \times 3 \times 2$

10 $m \times n$

11 $k \times k$

12 $p \div 3$

13 $2 \times x \times y \times 5$

14 $t + t - t$

YOU DO On your own: **Easy** Set B. Simplify.

6 QUESTIONS

1 $8 \times c \times b \times 3 \times a$

4 $3 \times y \times y$

2 $2 \times m \times w \times 10 \times w$

5 $a + a + b + b + b$

3 $k \times k \times k$

6 $4 \times d \div 2$

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Simplify.

a Simplify $5 \times a \times 2$.

Your teacher works this out. Copy every step.

b Simplify $8 \times c \times b \times 3 \times a$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Write in expanded form.

6 QUESTIONS

REMEMBER Leave out $\times$: $3 \times a = 3a$. Number first, letters in order.

1 $4mnk$

3 $9c^2$

5 a^3b

2 $\frac{6a}{b}$

4 $3xy^2$

6 $\frac{5m}{2n}$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.**Example 3** Write in expanded form.**a** Write $4mnk$ in expanded form.

Your teacher works this out. Copy every step.

b Write $9c^2$ and $\frac{6a}{b}$ in expanded form.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Patterns and rules.

4 QUESTIONS

REMEMBER $a \times a = a^2$; $a \div 5 = a/5$.**1** Write a rule for: $4 + 4 = 2 \times 4$, $7 + 7 = 2 \times 7$, $10 + 10 = 2 \times 10$ **3** Matchstick squares: 1 square uses 4 sticks, 2 squares use 7, 3 squares use 10. Find a rule for n squares.**2** Write a rule for: $3 \times 0 = 0$, $8 \times 0 = 0$, $15 \times 0 = 0$ **4** Is $3a^2$ the same as $(3a)^2$? Explain.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1Simplify $y + y + y$.**Level 2**Simplify $4 \times p \times 3 \times q$.**Level 3**Write $5a^2b$ in expanded form.

Extension · Stretch yourself**Worked example**

Write a rule for the pattern: $1 \times 3 = 2^2 - 1$, $2 \times 4 = 3^2 - 1$, $3 \times 5 = 4^2 - 1$

1. First number is n , second is $n + 2$
2. The square is of $n + 1$

Answer: $n(n + 2) = (n + 1)^2 - 1$

YOU DO

On your own: read the worked example, then try these.

- E1** Write a rule for: $5 \times 5 - 4 \times 6 = 1$, $8 \times 8 - 7 \times 9 = 1$, $3 \times 3 - 2 \times 4 = 1$

- E2** Test your rule with a number of your own.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. A **variable** is a letter that stands for a number.
 2. Leave out $\times$: $3 \times a = 3a$. Number first, letters in order.
 3. $a \times a = a^2$; $a \div 5 = a/5$.
 4. Expanded form shows every $\times$.

Level 1

Rule for $3 - (-3) = 2 \times 3 \dots$

1. Replace the number with a

Answer: $a - (-a) = 2a$

Level 2

$5 \times a \times 2$

1. $5 \times 2 = 10$

Answer: $10a$

Level 3

$2 \times m \times w \times 10 \times w$

1. $20 \times m \times w^2$

Answer: $20mw^2$

Extension

$9c^2$

1. $9 \times c \times c$

Answer: $9 \times c \times c$

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LESSON
5.04

Expressions from words

Date _____

Learning intentions

We are learning to:

- write algebraic expressions from worded statements

Success criteria

I can:

- ☐ know the words for +, −, × and ÷ (sum, difference, product, quotient)
- ☐ write an expression using a variable
- ☐ use brackets when needed

Key terms Write the definition as your teacher explains it.

Expression

Consecutive

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Write an expression using N .

a 3 times the number N

b the difference between N and 5

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Write an expression. Use N for the number. **12 QUESTIONS**

REMEMBER Sum $\rightarrow +$, difference $\rightarrow -$.

1 the number plus 7

2 3 times the number

3 the number minus 5

4 the number divided by 4

5 the next consecutive number

6 one-third of the number

Questions 7–12: write an expression.

7 5 more than x

8 x less 8

9 double x

10 x shared by 3

11 x squared

12 the product of x and y

YOU DO On your own: **Easy** Set B. Write an expression.

6 QUESTIONS

1 the sum of M and 3

4 the product of a and b

2 5 more than B

5 10 less than y

3 $2H$ decreased by k

⑥ p shared equally among 6

 Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Write an expression.

a the next consecutive number after N

Your teacher works this out. Copy every step.

b one-third of N

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Write an expression.

6 QUESTIONS

REMEMBER Product $\rightarrow \times$, quotient $\rightarrow \div$.

① double the sum of x and 4

[illegible]

③ 7 less than 3 times w

[illegible]

5 the number before n

[illegible]

② the square of t

④ half of the difference between m and 1

⑥ the average of a and b

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Write an expression.

a $2H$ decreased by k

Your teacher works this out. Copy every step.

b double the sum of x and 4

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Write an expression for each situation.

4 QUESTIONS

REMEMBER More than $\rightarrow +$, less than $\rightarrow -$ (order!).

1 The cost of n pens at \$3 each.

3 A rectangle is L cm long and 5 cm wide. Write its perimeter.

2 Sam is s years old. His sister is 4 years older. Find the sum of their ages.

4 The change from \$50 after buying k drinks at \$2.50 each.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Write “the number plus 12” using N .

Level 2

Write “4 less than y ”.

Level 3

Write “the product of m and 5, then add 2”.

 Extension · Stretch yourself Worked example

Write an expression for the sum of three consecutive numbers starting at n , and simplify it.

1. $n + (n + 1) + (n + 2)$
2. Collect like terms

Answer: $3n + 3$

YOU DO

On your own: read the worked example, then try these.

- E1** Write and simplify the sum of three consecutive even numbers starting at e .

- E2** Explain why the sum of 3 consecutive numbers is always divisible by 3.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Sum $\rightarrow +$, difference $\rightarrow -$.
 3. More than $\rightarrow +$, less than $\rightarrow -$ (order!).

2. Product $\rightarrow \times$, quotient $\rightarrow \div$.
4. Use brackets when a whole group is multiplied.

 Level 1

3 times N

1. $3 \times N$

Answer: $3N$

 Level 2

N decreased by 5

1. $N - 5$

Answer: $N - 5$

 Level 3

Double the sum of x and 4

1. Brackets first

Answer: $2(x + 4)$

 Extension

3 consecutive numbers

1. $n + n + 1 + n + 2$

Answer: $3n + 3$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.05

Substitution

Date _____

Learning intentions

We are learning to:

- find the value of an expression by substituting numbers

Success criteria

I can:

- ☐ replace each variable with its value, using brackets
- ☐ follow the order of operations
- ☐ substitute into a formula

Key terms Write the definition as your teacher explains it.

Substitute

Evaluate

Formula

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Substitute $a = 2$, $b = 5$ and $c = 6$.

a If $a = 2$ and $b = 5$, find $a + b$.

b If $a = 2$ and $c = 6$, find $3a - c$.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: If $a = 2$, $b = 5$ and $c = 6$, evaluate:

14 QUESTIONS

REMEMBER Write the expression.

1 $a + b$

2 $c - a$

3 $4b$

4 bc

5 $a + b + c$

6 $3a - c$

Questions 7–14: if $x = 3$ and $y = 4$, evaluate.

7 $x + y$

8 $2x$

9 xy

10 $y - x$

11 $5y$

12 x^2

13 $3x + y$

14 $10 - y$

YOU DO

On your own: **Easy** Set B. If $a = 2$, $b = 5$ and $c = 6$, evaluate:

6 QUESTIONS

1 $a + 2b - c$

4 $\frac{c}{a}$

2 abc

5 $2(a + b)$

3 c^2

6 $b^2 - c$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Substitute, then use the formula $C = 30h + 19$.

a If $a = 2$, $b = 5$, $c = 6$, find abc .

Your teacher works this out. Copy every step.

b If $C = 30h + 19$, find C when $h = 3$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. If $C = 30h + 19$, find C when:

6 QUESTIONS

REMEMBER Replace each variable with its value in brackets.

1 $h = 3$

3 $h = 5$

5 $h = 0$

2 $h = -1$

4 $h = 2.5$

6 $h = 10$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Substitute into $C = 30h + 19$.

a Find C when $h = -1$.

Your teacher works this out. Copy every step.

b Find C when $h = 2.5$.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Substitute into each formula.

4 QUESTIONS

REMEMBER Use the order of operations.

1 $P = 2L + 2W$. Find P when $L = 8$ and $W = 3$.

3 $F = 1.8C + 32$. Find F when $C = 25$.

2 $A = \frac{1}{2}bh$. Find A when $b = 10$ and $h = 7$.

4 If $x = -3$, evaluate $x^2 + 2x$.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

If $m = 4$, evaluate $5m$.

Level 2

If $x = 3$ and $y = 7$, evaluate $2x + y$.

Level 3

If $A = l^2 - 3l$, find A when $l = -2$.

Extension · Stretch yourself**Worked example**

A taxi charges $C = 4 + 2.5k$ dollars for k km. How far can you go for \$39?

1. $39 = 4 + 2.5k$
2. $35 = 2.5k$
3. $k = 35 \div 2.5$

Answer: 14 km

YOU DO

On your own: read the worked example, then try these.

- E1** Using the same formula, find the cost of a 22 km trip.

- E2** A phone plan costs $C = 15 + 0.2m$ for m minutes. How many minutes for \$27?

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Write the expression.
 2. Replace each variable with its value in brackets.
 3. Use the order of operations.
 4. Remember $3a$ means $3 \times a$.

Level 1

$$a + b, a = 2, b = 5$$

1. $2 + 5$

Answer: 7

Level 2

$$3a - c$$

1. $3 \times 2 - 6$

Answer: 0

Level 3

$$C = 30h + 19, h = -1$$

1. $30 \times (-1) + 19$

Answer: -11

Extension

Taxi for \$39

1. $2.5k = 35$

Answer: 14 km

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.07

Adding and subtracting terms

Date _____

Learning intentions

We are learning to:

- simplify expressions by collecting like terms

Success criteria

I can:

- ☐ identify **like terms** (same letters, same powers)
- ☐ add and subtract the coefficients of like terms
- ☐ collect several sets of like terms

Key terms Write the definition as your teacher explains it.**Like terms****Coefficient**

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO

With your teacher: copy what they write on the board.

Example 1

Simplify by collecting like terms.

a $3m + 4m$

b $12m - 4m$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** Set A. Questions 1–6: Simplify.

14 QUESTIONS

REMEMBER **LIKE TERMS** have exactly the same letters and powers.

1 $3m + 4m$

2 $12m - 4m$

3 $5a + a$

4 $9k - 9k$

5 $2x + 3x + 4x$

6 $7y - 4y - y$

Questions 7–14: simplify.

7 $4a + 3a$

8 $9x - 5x$

9 $2m + m$

10 $6k - k$

11 $3p + 2q + p$

12 $8y - 3y + y$

13 $5ab + 2ab$

14 $7c - 7c$

YOU DO

On your own: **Easy** Set B. Are these like terms? Write yes or no.

6 QUESTIONS

1 $3a$ and $5a$

4 m^2 and $3m$

2 $2x$ and $2y$

5 6 and $6p$

3 $4ab$ and $7ba$

6 $-2k^2$ and k^2



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Simplify by collecting like terms.

a $7y - 4y - y$

Your teacher works this out. Copy every step.

b $6uv + 4 + 12uv$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Simplify.

6 QUESTIONS

REMEMBER Add or subtract only like terms.

1 $6uv + 4 + 12uv$

3 $8x - 3 + 2x + 7$

5 $10 - 3m - 4 + 5m$

2 $5a + 3b + 2a + b$

4 $4pq - pq + 3qp$

6 $2y + 5 - 2y$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Simplify. Collect each kind of term separately.

a $5h^2 + 6h + 8h^2 + 11h$

b $13h^3 - 6h - 8h - 11h^3$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Simplify.

4 QUESTIONS

REMEMBER Keep the sign in front of each term.

1 $5h^2 + 6h + 8h^2 + 11h$

3 $3ab - 2a + 5ab + 7a - b$

2 $13h^3 - 6h - 8h - 11h^3$

4 A triangle has sides $2x$, $3x + 1$ and $x + 4$. Write and simplify its perimeter.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill** Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Simplify $6k + 3k$.

Level 2

Simplify $4a + 7 - a + 2$.

Level 3

Simplify $3x^2 + 5x - x^2 - 8x$.

Extension · Stretch yourself**Worked example**Simplify $4x^2y - 3xy^2 + x^2y + 5xy^2 - xy$.

1. Like terms: $4x^2y + x^2y$
2. $-3xy^2 + 5xy^2$
3. $-xy$ has no partner

Answer: $5x^2y + 2xy^2 - xy$ **YOU DO**

On your own: read the worked example, then try these.

- E1**
- Simplify
- $7a^2 - 3a + 2 - a^2 + 5a - 8$
- .

- E2**
- Find the missing term:
- $5m + ? - 2m = 9m$

Lesson summary

Missed the lesson? Start here.

Remember

1. Like terms have exactly the same letters and powers.
2. Add or subtract only like terms.
3. Keep the sign in front of each term.
4. Write the answer with the highest power first.

Level 1

$3m + 4m$

1. $3 + 4 = 7$

Answer: $7m$ **Level 2**

$7y - 4y - y$

1. $7 - 4 - 1$

Answer: $2y$ **Level 3**

$6uv + 4 + 12uv$

1. $6 + 12 = 18$

Answer: $18uv + 4$ **Extension**

$5h^2 + 6h + 8h^2 + 11h$

1. h^2 terms, then h terms

Answer: $13h^2 + 17h$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.08

Multiplying terms

Date _____

Learning intentions

We are learning to:

- multiply algebraic terms

Success criteria

I can:

- ☐ multiply the numbers first, then the letters
- ☐ write repeated letters as powers
- ☐ use the sign rules for negative terms

Key terms Write the definition as your teacher explains it.

Coefficient

Power

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Simplify.

a $3m \times 4m$

Your teacher works this out. Copy every step.

b $a \times a \times a \times a$

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Simplify.

14 QUESTIONS

REMEMBER Multiply the numbers (coefficients).

1 $3 \times 4m$

2 $2a \times 5$

3 $a \times b$

4 $a \times a \times a \times a$

5 $x \times 7 \times y$

6 $3m \times 4m$

Questions 7–14: simplify.

7 $5 \times 2x$

8 $3a \times 4$

9 $x \times y \times 2$

10 $m \times m$

11 $2p \times 3q$

12 $4k \times k$

13 $-3 \times 5y$

14 $-2a \times (-4b)$

YOU DO

On your own: **Easy** Set B. Simplify.

6 QUESTIONS

1 $6p \times 2q$

4 $5ab \times 2a$

2 $9v \times 4 \times 5v$

5 $4x \times 3x \times x$

3 $k \times k \times 2$

6 $2c \times 2c \times 2c$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Simplify.

a $9v \times 4 \times 5v$

Your teacher works this out. Copy every step.

b $3pq \times 5pq$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Simplify. Watch the signs.

6 QUESTIONS

REMEMBER Multiply the letters, in alphabetical order.

1 $-7d \times (-3d)$

3 $3x \times (-x)$

5 $-3x^2y \times 2x$

2 $-2a \times 6b$

4 $-4m \times (-m) \times 2$

6 $5 \times (-t) \times (-t)$



Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.**Example 3** Simplify. Watch the signs.

a $-7d \times (-3d)$

Your teacher works this out. Copy every step.

b $lmn \times lmn \times lmn$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Simplify.

4 QUESTIONS

REMEMBER Same letter repeated → use a power.

1 $3pq \times 5pq$

3 $2a^2 \times 3a^3$

2 $lmn \times lmn \times lmn$

4 A rectangle is $3x$ cm long and $2y$ cm wide. Write its area.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skillChoose **ONE** question. Show your working.

Name _____ Class _____

Level 1Simplify $5 \times 3k$.**Level 2**Simplify $4a \times 6ab$.**Level 3**Simplify $-2x \times 3x \times (-y)$.

Extension · Stretch yourself**Worked example**Simplify $(2ab)^3$.

- $2ab \times 2ab \times 2ab$
- Numbers: $2 \times 2 \times 2 = 8$
- a^3b^3

Answer: $8a^3b^3$ **YOU DO**

On your own: read the worked example, then try these.

E1 Simplify $(-3m)^2$.**E2** Find the missing term: $4x \times ? = 20x^2y$.**Lesson summary**

Missed the lesson? Start here.

Remember

1. Multiply the numbers (coefficients).
3. Same letter repeated $\rightarrow$ use a power.

2. Multiply the letters, in alphabetical order.
4. Signs: same $\rightarrow$ +, different $\rightarrow$ -.

Level 1 $3m \times 4m$

1. $3 \times 4 = 12$, $m \times m = m^2$

Answer: $12m^2$ **Level 2** $9v \times 4 \times 5v$

1. $9 \times 4 \times 5 = 180$

Answer: $180v^2$ **Level 3** $-7d \times (-3d)$

1. $- \times - = +$

Answer: $21d^2$ **Extension** $(2ab)^3$

1. $8, a^3, b^3$

Answer: $8a^3b^3$

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LESSON
5.09

Dividing terms

Date _____

Learning intentions

We are learning to:

- divide algebraic terms

Success criteria

I can:

- ☐ write a division as a fraction
- ☐ cancel common factors in the numbers and the letters
- ☐ simplify answers that are still fractions

Key terms Write the definition as your teacher explains it.

Quotient

Cancel

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Simplify.

a $15m \div 3$

b $9v \div v$

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Simplify.

12 QUESTIONS

REMEMBER Write the division as a fraction.

1 $15m \div 3$

2 $9v \div v$

3 $\frac{49y}{7}$

4 $8ab \div 2$

5 $20k \div 5k$

6 $\frac{6pq}{q}$

Questions 7–12: simplify.

7 $12a \div 4$

8 $18m \div 6$

9 $10x \div x$

10 $24pq \div 8p$

11 $6ab \div 3b$

12 $15k \div 5k$

YOU DO

On your own: **Easy** Set B. Simplify.

6 QUESTIONS

1 $\frac{12x^2y}{4xy}$

4 $24t^3 \div 8t$

2 $18m^2 \div 6m$

5 $\frac{3x}{12}$

3 $\frac{10ab}{5b}$

6 $\frac{4a}{6ab}$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Simplify. Write the division as a fraction.

a $\frac{49y}{7}$

Your teacher works this out. Copy every step.

b $\frac{12x^2y}{4xy}$

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Simplify.

6 QUESTIONS

REMEMBER Cancel the numbers by their HCF.

1 $32de^2 \div 10de$

3 $\frac{-15x^2}{-5x}$

5 $\frac{14pq^2}{21pq}$

2 $\frac{-20mn}{4m}$

4 $\frac{9a^2b^2}{3ab}$

6 $-36y^3 \div 4y^2$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Simplify.

a $32de^2 \div 10de$

Your teacher works this out. Copy every step.

b $\frac{-3 \times 2 y \times 2 x}{3 x \times (-x)}$

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Simplify.

4 QUESTIONS

REMEMBER Cancel letters that are on top and bottom.

1 $\frac{-3 \times 2 y \times 2 x}{3 x \times (-x)}$

3 $\frac{10 m 2 n}{2 m \times 5 n}$

2 $\frac{4 a \times 6 b}{8 a b}$

4 A rectangle has area $24xy \text{ cm}^2$ and length $6x \text{ cm}$. Find its width.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Simplify $21p \div 7$.

Level 2

Simplify $\frac{16ab^2}{4b}$.

Level 3

Simplify $\frac{-10 \times 3}{15x}$.

Extension · Stretch yourself

Worked example

Simplify $\frac{6a3b \times 4ab2}{12a2b2}$.

1. Top: $24a^4b^3$

2. $\div 12a^2b^2$

Answer: $2a^2b$

YOU DO

On your own: read the worked example, then try these.

E1 Simplify $\frac{(3m)2}{6m}$.

E2 Simplify $\frac{8 \times 2y \times 3y}{6xy \times 2x}$.

Lesson summary

Missed the lesson? Start here.

Remember 1. Write the division as a fraction.
3. Cancel letters that are on top and bottom.

2. Cancel the numbers by their HCF.
4. Signs: same $\rightarrow +$, different $\rightarrow -$.

Level 1

$15m \div 3$

1. $15 \div 3 = 5$

Answer: $5m$

Level 2

$\frac{12 \times 2y}{4xy}$

1. $12 \div 4 = 3$, cancel x and y

Answer: $3x$

Level 3

$32de^2 \div 10de$

1. $32/10 = 16/5$, cancel d and e

Answer: $\frac{16e}{5}$

Extension

$\frac{6a3b \times 4ab2}{12a2b2}$

1. $24 \div 12 = 2$

Answer: $2a^2b$

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